

从 Pods 到 Prompts:

Kubernetes 网络演进与 AI 新时代

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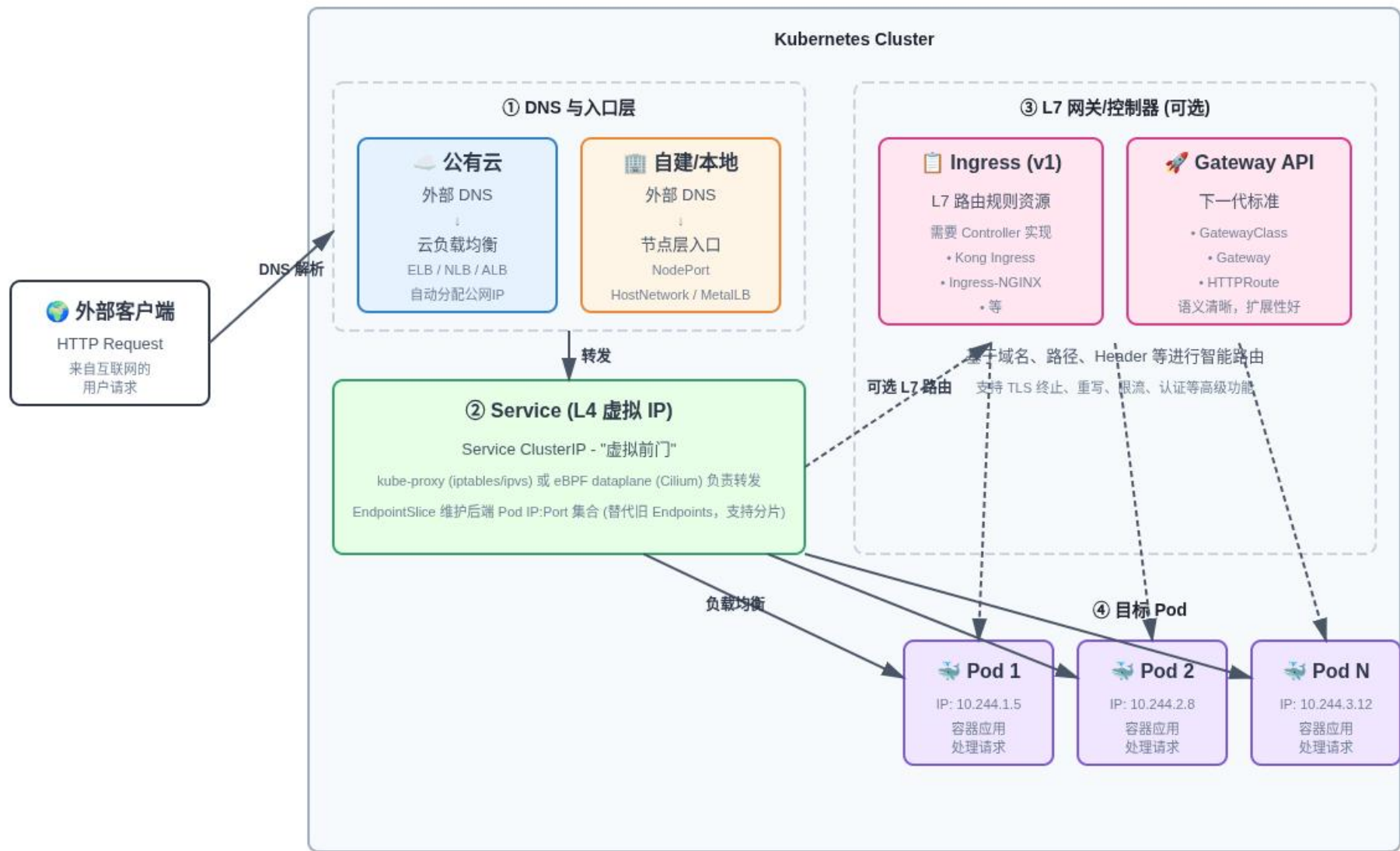
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Kubernetes的网络模型



从 Ingress 到 Gateway API

Ingress NGINX Retirement: What You Need to Know

By **Tabitha Sable (Kubernetes SRC)** | Tuesday, November 11, 2025

To prioritize the safety and security of the ecosystem, Kubernetes SIG Network and the Security Response Committee are announcing the upcoming retirement of [Ingress NGINX](#). Best-effort maintenance will continue until March 2026. Afterward, there will be no further releases, no bugfixes, and no updates to resolve any security vulnerabilities that may be discovered.

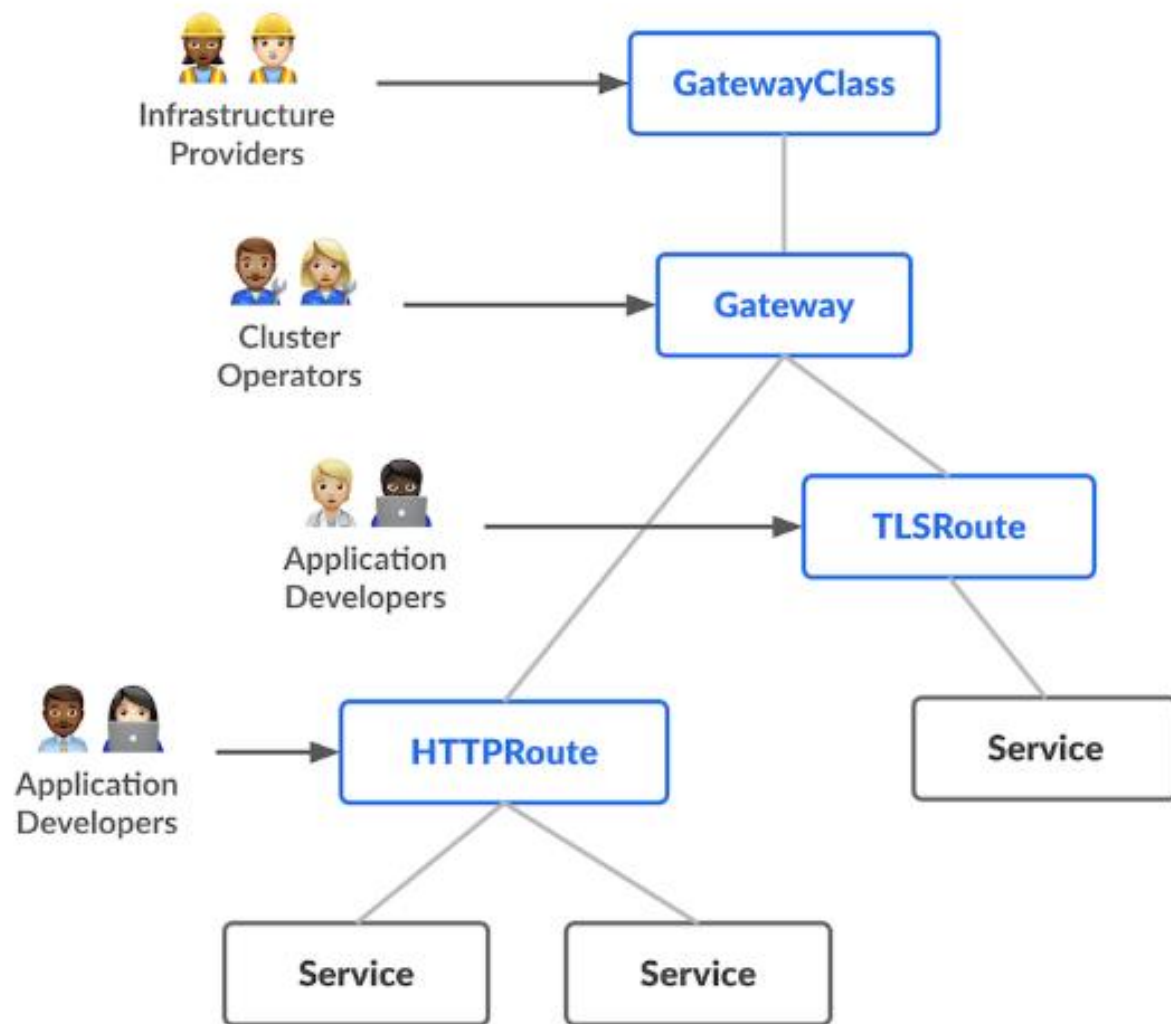
Existing deployments of Ingress NGINX will continue to function and installation artifacts will remain available.

We recommend migrating to one of the many alternatives. Consider [migrating to Gateway API](#), the modern replacement for Ingress. If you must continue using Ingress, many alternative Ingress controllers are [listed in the Kubernetes documentation](#). Continue reading for further information about the history and current state of Ingress NGINX, as well as next steps.

About Ingress NGINX

[Ingress](#) is the original user-friendly way to direct network traffic to workloads running on Kubernetes. ([Gateway API](#) is a newer way to achieve many of the same goals.) In order for an Ingress to work in your cluster, there must be an [Ingress controller](#) running. There are many Ingress controller choices available, which serve the needs of different users and use cases. Some are cloud-provider specific, while others have more general applicability.

[Ingress NGINX](#) was an Ingress controller, developed early in the history of the Kubernetes project as an example implementation of the API. It became very popular due to its tremendous flexibility, breadth of features, and independence from any particular cloud or infrastructure provider. Since those days, many other Ingress controllers have been created within the Kubernetes project by community groups, and by cloud native vendors. Ingress NGINX has continued to be one of the most popular, deployed as part of many hosted Kubernetes platforms and within innumerable independent users' clusters.




```
kind: HTTPRoute
apiVersion: gateway.networking.k8s.io/v1beta1
metadata:
  name: orders-httproute
spec:
  parentRefs:
  - group: gateway.networking.k8s.io
    kind: Gateway
    name: kong-gateway
  hostnames:
  - orders.example.com
  rules:
  - matches:
    - path:
        type: PathPrefix
        value: /orders
    backendRefs:
    - name: orders
      port: 8000
```

Kubernetes Cluster

KIC (Kong Ingress Controller)

Kong Ingress Controller

监听 K8s 资源 → Kong 配置

🏠 处理的资源:

标准 Ingress

K8s Ingress 规则

Kong CRDs

KongPlugin
KongConsumer

Gateway API
(部分支持)

🏠 生成配置:

Kong Gateway 配置

- Routes (路由)
- Services (服务)
- Plugins (插件)
- Consumers (消费者)

协作

KO (Kong Operator)

Kong Operator

网关生命周期 & 运维抽象

🔧 核心 CRDs:

ControlPlane

(v2beta1)

控制器 & 功能开关
GW API 集成

DataPlane

(v1beta1)

数据面部署
镜像/资源/探针

GatewayConfiguration

(v2beta1)

与 GatewayClass 绑定

KongPluginInstallation

(v1alpha1)

分发自定义插件

AIGateway

(v1alpha1)

AI 高层抽象

🏠 深度集成:

GatewayClass / Gateway / HTTPRoute

AI 时代与Kong AI Gateway

从 API 到 AI: 新的挑战

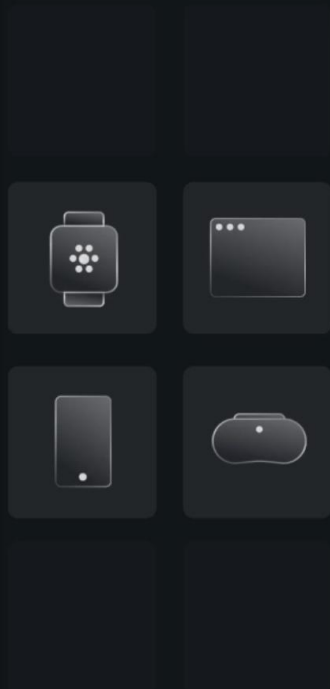
传统 API 时代

- 治理单位: API 请求
- 计量单位: 请求数量 (QPS)
- 成本模型: 按请求计费
- 统一接口: RESTful / GraphQL
- 安全重点: 认证授权、参数校验

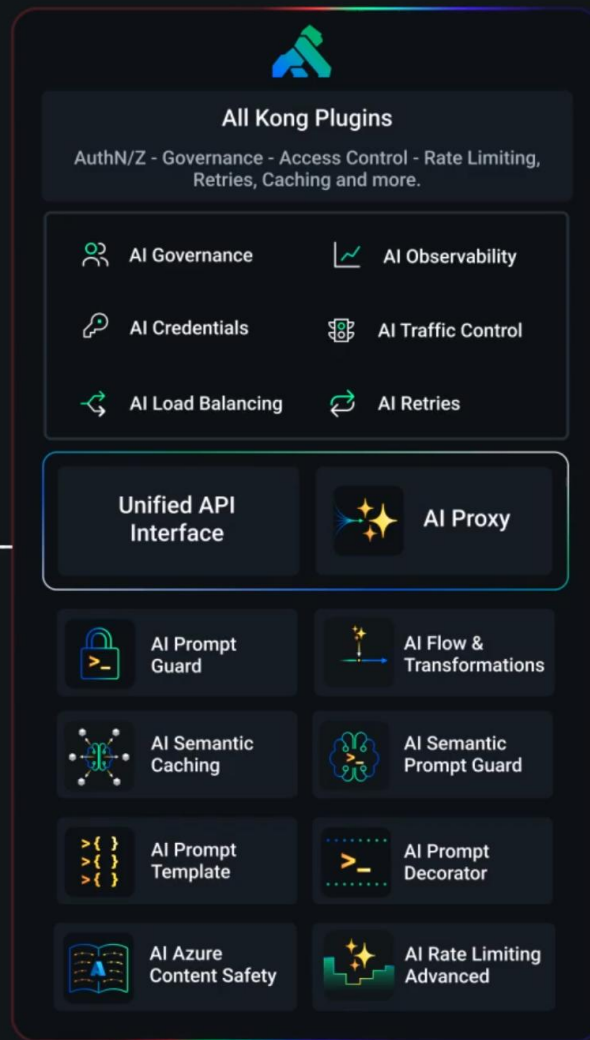
NEW AI 时代

- 治理单位: LLM 调用
- 计量单位: Token / 上下文成本 / 模型类型
- 成本模型: Token 计费 (输入+输出)
- 多厂商接口: 各有差异需归一化
- 安全重点: Prompt 注入、内容安全、数据泄露
- 新需求: 配额管理、成本预算、语义缓存

Apps & Agents

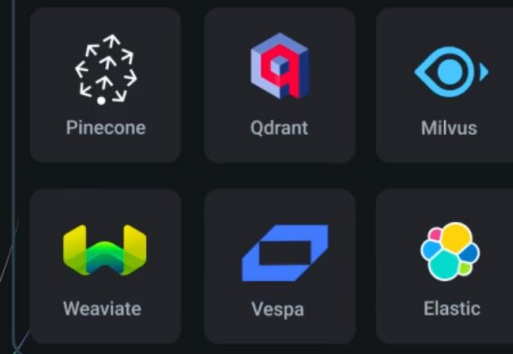


AI Gateway

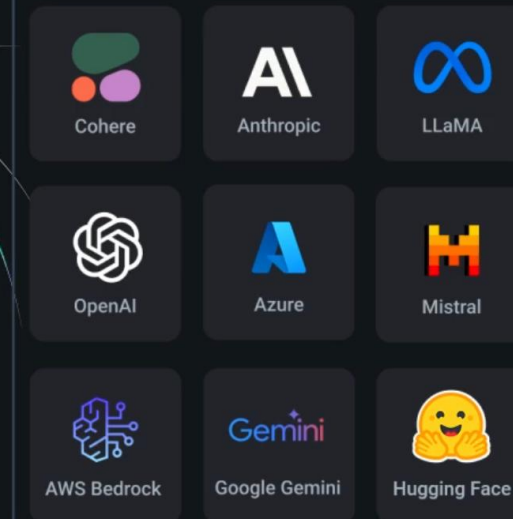


AI Providers

Vector DBs



LLMs





统一 LLM 接入

将不同厂商 API 归一化处理, 支持统一路由配置、模型提供商切换

插件: *ai-proxy, ai-proxy-advanced*



Token 维度治理

按 Token 数量限速、响应成本控制、多时间窗口策略、Redis 分布式支持

插件: *ai-rate-limiting-advanced*



Prompt 安全与治理

防止 Prompt 注入、内容安全检测、模板管理、请求装饰与增强

插件: *ai-prompt-guard, ai-prompt-decorator, ai-prompt-template*



请求/响应增强

请求/响应转换、语义缓存、响应内容安全检测、智能降级

插件: *ai-request-transformer, ai-response-transformer, ai-semantic-cache*



可观测性与成本核算

LLM 调用统计、Token 使用追踪、成本核算报表、异常告警

结合网关指标、日志、Analytics 平台



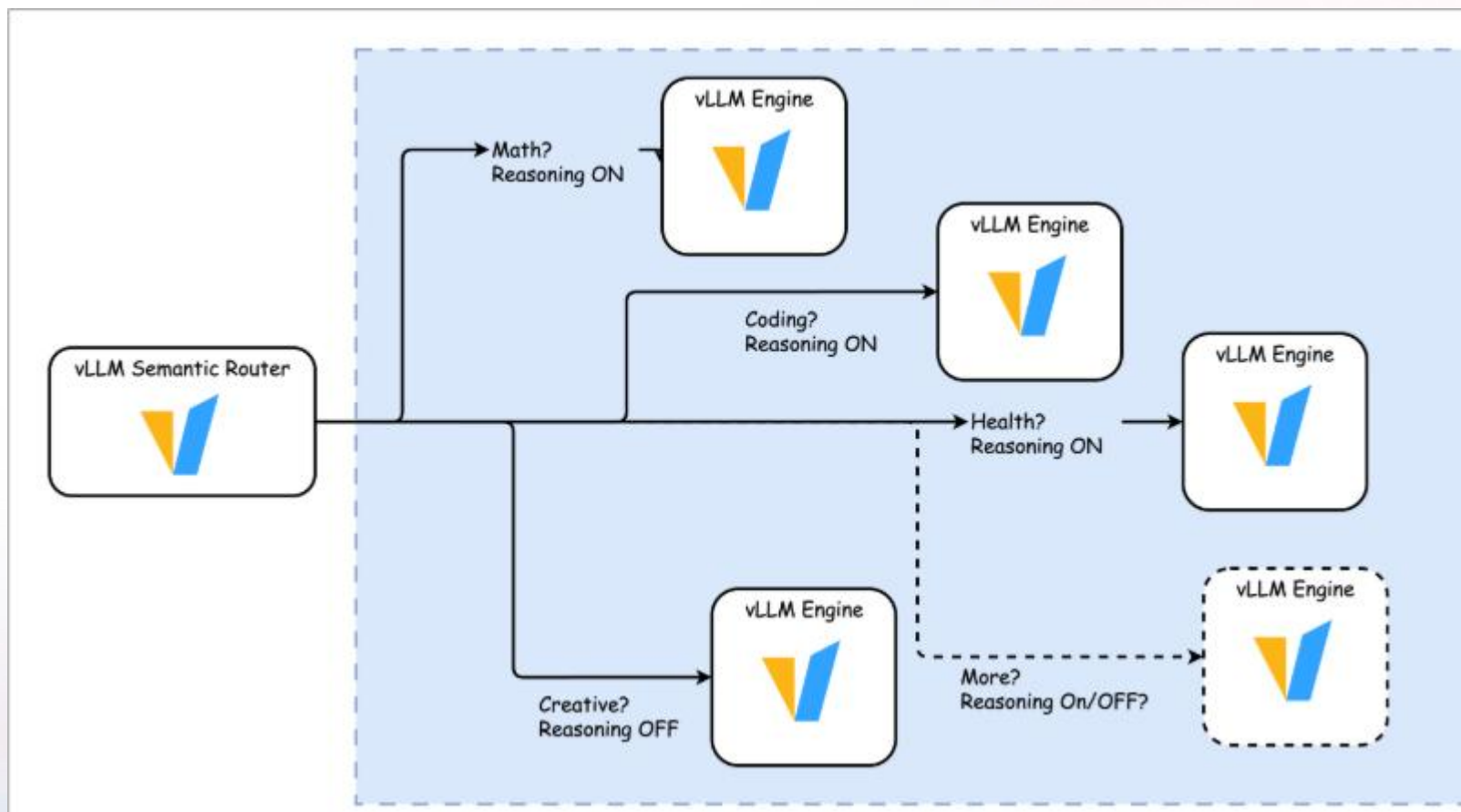
认证、授权与配额

统一身份认证、细粒度授权、配额管理、多租户隔离

插件: *key-auth, oauth2, acl, quota-management*



AI + Cloud Native



Kong **Konnect**





CHINA
OpenInfra Days

